Sumit Togarshi

Student

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Seeking to leverage my strong problem-solving skills and passion for building efficient software solutions to alleviate development bottlenecks and support agile feature delivery in the Software Engineering Intern position . I'm excited about the opportunity to deepen my understanding of of real-world user needs and technical challenges within the software development industry. As of today, I am ready to contribute as a Software Engineering Intern .

Projects

Smart Car Using Arduino | Personal Team Project

- Built a prototype smart car using an Arduino microcontroller, integrating sensors for movement and obstacle detection.
- Programmed in C/C++ using the Arduino IDE; practiced core embedded programming and real-time control logic.
- Focused on hands-on learning with electronics, circuit design, and control algorithms

OpsPilot:Multi-Agent Orchestration for Legacy Modernization

- Engineered the OpsPilot multi-agent framework utilizing AutoGen's conversation programming to autonomously modernize legacy enterprise codebases (COBOL, C, PL/SQL) into cloud-native architectures.
- Designed a deterministic three-phase lifecycle (Strategy, Execution, Verification) featuring an adversarial self-refine loop and structured Agent-to-Agent (A2A) message passing to ensure semantic fidelity without overlapping task ownership.
- Synthesized empirical research demonstrating the distributed architecture achieves 93% logic retention accuracy, outperforming single-agent LLM models by 18 percentage points while reducing migration time by an estimated 50%.

Traffic Flow Analysis

- Real-Time Detection Pipeline: Engineered a production-grade vehicle detection and traffic counting system using YOLOv8 and ByteTrack to track specific vehicle classes across defined zones..
- Architecture & Optimization: Resolved 15 critical bugs from a baseline implementation, introducing resolution-independent polygon scaling, dynamic frame skipping, and robust CLI/Tkinter GUI access.
- **Automated Analytics:** Designed automated data pipelines to export JSON vehicle summaries and CSV transition matrices, alongside a live video HUD displaying FPS and bounding boxes.

Education

HKBK College of Engineering | Bachelor of Engineering in Computer Science Expected *Graduation:2027*
Relevant Coursework: Intro to Machine Learning, Data Structures & Algorithm, Computer Network, Cloud Computing
Licenses & Certifications: Stanford Online – Supervised Machine Learning, Oracle cloud Infrastructure Certification
CGPA: 7.83

Skills

- Programming & Tools: Python, SQL, C/C++, Java, Git, Linux(basics)
- Machine Learning Frameworks: PyTorch, pandas, NumPy
- Data Visualisation: Matplotlib, Seaborn, Excel, MATLAB
- Deployment & Cloud: Streamlit Cloud, AWS Cloud (EC2, S3, VPC)
- C & C++(OOP's)
- HTML & CSS